

C++ PROGRAMMING
DSA0108
GALI GIRI
192472404

1)STUDENT:

```
1 #include <iostream>
2 using namespace std;
3 class Student {
4     string name;
5     int rollNo;
6 public:
7     Student(string n, int r) {
8         name = n;
9         rollNo = r;
10        cout << "Constructor called for " << name << " (Roll: " << rollNo << ")\n";
11    }
12    ~Student() {
13        cout << "Destructor called for " << name << " (Roll: " << rollNo << ")\n";
14    }
15    void display() {
16        cout << "Student Name: " << name << "\n";
17        cout << "Roll Number: " << rollNo << "\n";
18    }
19 };
20 int main() {
```

OUTPUT:

OUTPUT

```
Constructor called for gali giri (Roll: 192472404)
Student Name: gali giri
Roll Number: 192472404
Destructor called for gali giri (Roll: 192472404)
```

2)BOOK:

```
1 #include <iostream>
2 #include <cstring>
3 using namespace std;
4
5 class Book {
6     char title[100];
7     char author[100];
8
9 public:
10    // Constructor
11    Book(char t[], char a[]) {
12        strcpy(title, t);
13        strcpy(author, a);
14        cout << "Constructor called: Book '" << title << "' created." << endl;
15    }
16
17    // Destructor
18    ~Book() {
19        cout << "Destructor called: Book '" << title << "' destroyed." << endl;
```

OUTPUT:

OUTPUT

```
Constructor called: Book 'C++ Programming' created.
Title: C++ Programming
Author: RAMANUJAM
Destructor called: Book 'C++ Programming' destroyed.
```

3)RECTANGLE:

```
1 #include <iostream>
2 using namespace std;
3
4 class Rectangle {
5 private:
6     double length;
7     double breadth;
8 public:
9     Rectangle(double l, double b) {
10         length = l;
11         breadth = b;
12         cout << "Rectangle created (Constructor called).\n";
13     }
14     ~Rectangle() {
15         cout << "Rectangle destroyed (Destructor called).\n";
16     }
17     double area() {
18         return length * breadth;
19     }
20 };
21
```

OUTPUT:

OUTPUT

```
Enter length and breadth: Rectangle created (Constructor called).
Area = 200
Rectangle destroyed (Destructor called).
```

4)CAR:

```
1 #include <iostream>
2 #include <string>
3
4 class Car {
5 public:
6     Car() {
7         std::cout << "Constructor called: A new car has been created!" << std::endl;
8     }
9     ~Car() {
10         std::cout << "Destructor called: The car has been destroyed." << std::endl;
11     }
12     void displayMessage() {
13         std::cout << "The car is ready to drive!" << std::endl;
14     }
15 };
16 int main() {
17     Car myCar;
18     myCar.displayMessage();
19     return 0;
20 }
```

OUTPUT:

OUTPUT

```
Constructor called: A new car has been created!
The car is ready to drive!
Destructor called: The car has been destroyed.
```

5)BANK:

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4 class BankAccount {
5 private:
6     string accountHolder;
7     double balance;
8 public:
9     BankAccount(string name, double initialBalance) {
10         accountHolder = name;
11         balance = initialBalance;
12         cout << "Account created for " << accountHolder << " with balance: $" <<
13     }
14     ~BankAccount() {
15         cout << "Account for " << accountHolder << " is being closed. Goodbye!"
16     }
17     void displayBalance() {
18         cout << "Current Balance: $" << balance << endl;
19     }
```

OUTPUT:

OUTPUT

```
Account created for GALI GIRI with balance: $50000.8
Current Balance: $50000.8
Account for GALI GIRI is being closed. Goodbye!
```